ANALYSIS REPORT

Wave 5.2R Stage 11 Uncertainty And Physics-Trust Calibration Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

Stage 11 completed all ten calibration entries without campaign failure. The frozen K01 curve remained the prediction center throughout. The campaign tested whether causal operating-support, analytical-disagreement, dense-model-disagreement, and five-seed ensemble signals could localize K01 error and support non-vacuous empirical intervals.

The qualified Stage 11 trust component is none. The strongest diagnostic candidate is `D01` with Spearman correlation `0.281`, top-quintile average precision `0.499`, and high-error capture `0.400`. No result changes K01 promotion status or authorizes Wave 6.

Scope And Leakage Boundary

- Dataset: polished dataset, setpoint inputs, forward surface only.
- Split: frozen `675/194/97` grouped split.
- Mean prediction: primary Stage 9 K01, unchanged.
- Calibration: validation partition only.
- Final evaluation: one held-out test pass.
- Runtime target-derived inputs: zero.
- Ensemble: five deterministic K01 seeds with identical architecture and optimization rules.

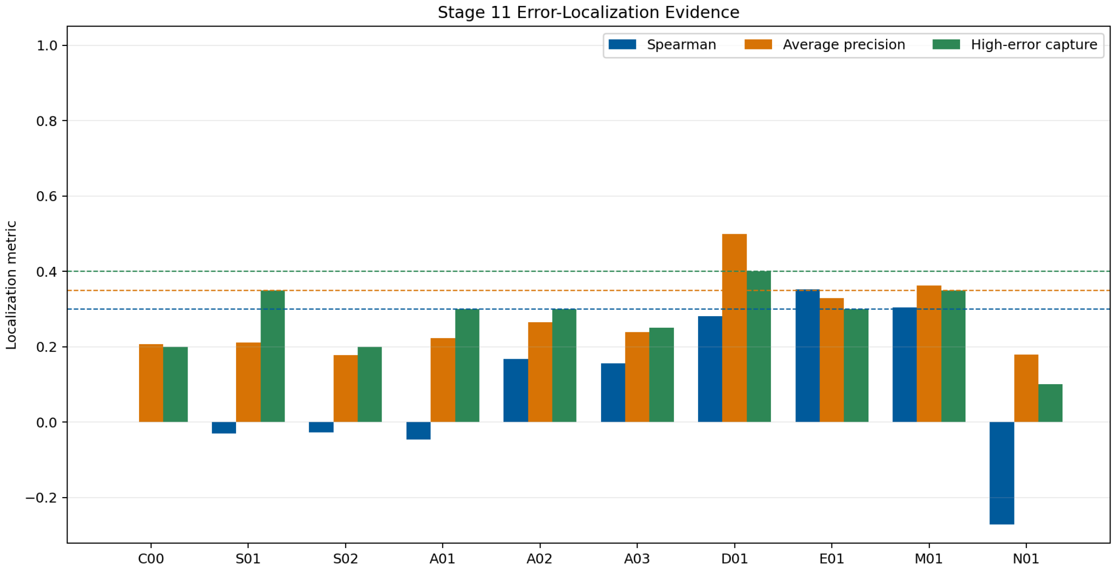
Candidate Results

Candidate labels are: `C00` constant control, `S01` condition distance, `S02` support boundary, `A01` PF-A/H04 disagreement, `A02` H04/K01 disagreement, `A03` PF-A/K01 disagreement, `D01` R00/K01 disagreement, `E01` five-seed spread, `M01` composite trust, and `N01` shuffled control.

ID	Spearman	AP	Capture	MAE@80	Cov90	Width90
`C00`	0.000	0.206	0.200	0.001372	0.907	0.006831
`S01`	-0.031	0.211	0.350	0.001189	0.907	0.006831
`S02`	-0.028	0.177	0.200	0.001365	0.896	0.006793
`A01`	-0.047	0.223	0.300	0.001271	0.904	0.006929
`A02`	0.167	0.265	0.300	0.001331	0.891	0.006208

Residual, Ensemble, And Composite Signals

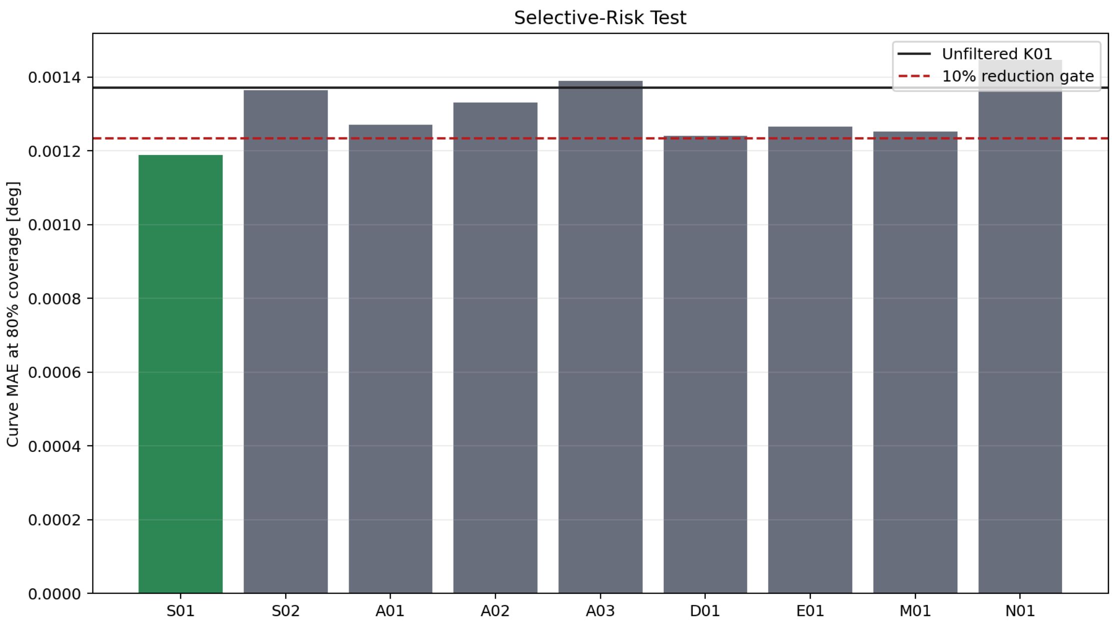
ID	Spearman	AP	Capture	MAE@80	Cov90	Width90
A03	0.156	0.239	0.250	0.001389	0.894	0.006500
D01	0.281	0.499	0.400	0.001241	0.907	0.006693
E01	0.352	0.328	0.300	0.001266	0.901	0.006521
M01	0.304	0.362	0.350	0.001252	0.898	0.006616
N01	-0.272	0.179	0.100	0.001446	0.897	0.006804



Stage 11 localization metrics

Error Localization

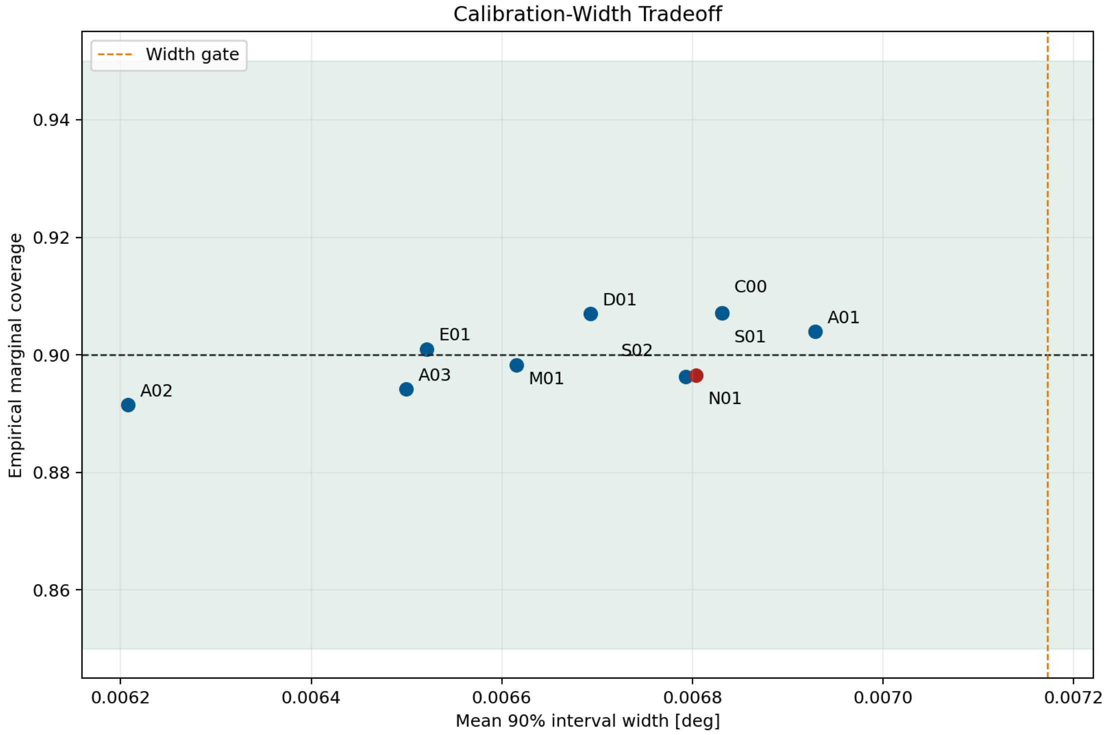
The primary question is whether uncertainty ranks actual held-out curve error. The constant control has no meaningful localization by construction, while the shuffled control tests whether the observed score distribution alone can reproduce the result. A candidate must exceed both controls and pass the fixed rank, average-precision, capture, and selective-risk gates.



Stage 11 selective-risk test

Interval Calibration

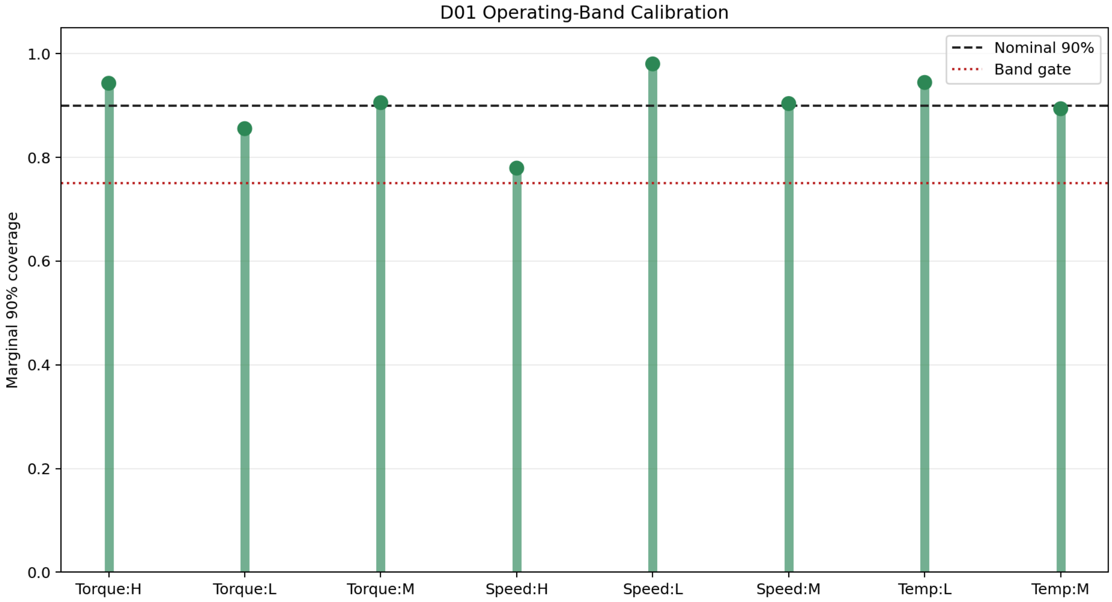
All intervals are centered on the frozen K01 prediction. Validation absolute residuals determine split-conformal quantiles; test labels do not tune widths. The fixed gate requires empirical 90-percent marginal coverage between `0.85` and `0.95` with mean width no more than `1.05` times the constant conformal control.



Stage 11 calibration-width tradeoff

Operating-Band Evidence

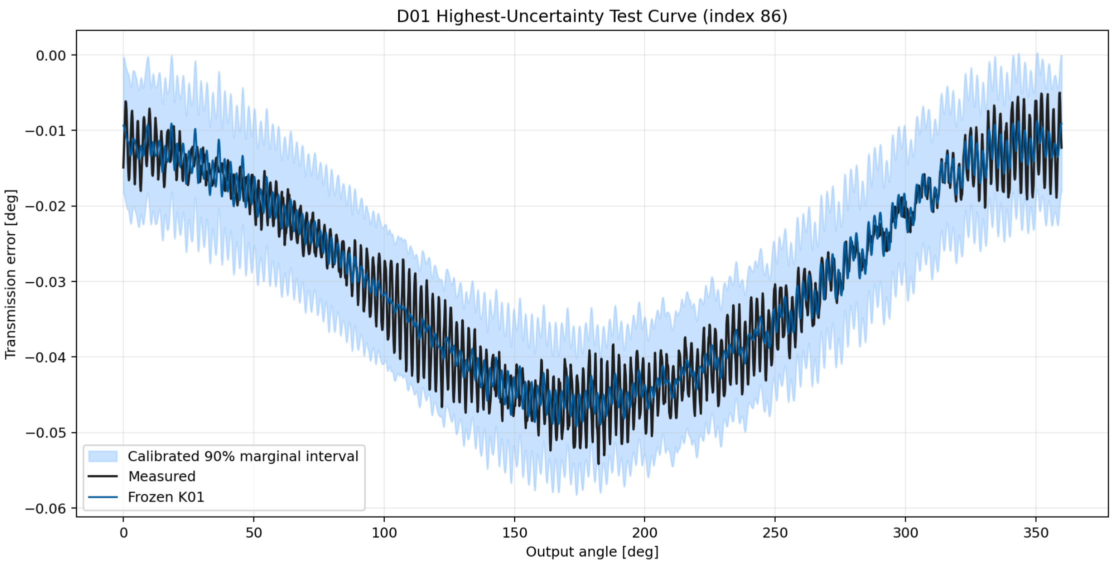
Torque, speed, and temperature bands use train-defined terciles. Any populated band with at least ten test curves must retain at least `0.75` marginal coverage. The Stage 3 support tier remains visible separately because only a small number of test curves occupy sparse or extrapolation tiers.



Stage 11 operating-band coverage

Representative Calibrated Curve

The plot below shows the highest-uncertainty test curve for `D01`. The interval is an empirical error band, not a mechanistic probability distribution of reducer TE.



Stage 11 representative interval

Deployment Cost

Simple condition or disagreement signals retain one primary K01 checkpoint. The ensemble candidate requires five K01 checkpoints and is therefore eligible only as offline research evidence unless a future single-pass trust head matches its calibration. The deployment gate remains a maximum `1.25` times the primary K01 checkpoint cost.

Decision

- Stage 11 status: `completed_without_qualified_trust_component`.
- Qualified trust component: none.
- Diagnostic best candidate: `D01`.
- Official mean prediction changed: no.
- K01 promoted: no.
- Physics-integrated Wave 6 authorized: no.
- Next step: Stage 12 advanced constraint optimization, applied only to ingredients that already showed isolated signal.

Reproducibility Evidence

- Campaign leaderboard: `output/training_campaigns/2026-07-29-20-49-33_wave52r_stage11_uncertainty_trust_calibration_2026_07_29/campaign_leaderboard.yaml`
- Gate summary: `output/training_campaigns/2026-07-29-20-49-33_wave52r_stage11_uncertainty_trust_calibration_2026_07_29/campaign_first_screen_gate_summary.yaml`
- Exit-gate summary: `output/analysis/wave_5_2r/stage11_uncertainty_physics_trust_calibration/closeout/stage11_exit_gate_summary.yaml`
- Preflight: `output/analysis/wave_5_2r/stage11_uncertainty_physics_trust_calibration/stage11_preflight_validation_summary.yaml`